Akoizo Original Practice - Form 2

Division C | Electric Vehicle | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which part of the vehicle is used to measure Vehicle Distance?

Unit: Planning the path: distances, geometry, and the reverse.
Worked-case data: Target Distance 7.00 m, Bottle Line Distance 2.00 m, vehicle length 50 cm, bottle 6.5 cm, desired margin 3 cm. How far does the MP travel forward and backward?
Separate investigation observations:
Setup: Target 8.00 m, Bottle Line 1.00 m, vehicle length 45 cm.
Run: MP forward travel programmed as 8.60 m.
Result: Bottle ended 1 cm short of the line.

A. The front axle

B. The center of mass

C. The Measurement Point at the rear

D. The bottle pusher

Answer: _____

2. (2 points) How must the vehicle be started?

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

A. With an unsharpened #2 pencil using a vertical motion

B. By hand push

C. With a phone app

D. By releasing a held wheel

Answer: _____

3. (2 points) What happens if the vehicle tips the bottle over while pushing?

Unit: The bottle pusher: catching, pushing, and the narrow-opening bonus.
Worked-case data: Compare pushers with openings of 9.0 cm and 15.0 cm. The 9.0 cm version catches the bottle in 8 of 10 tests; the 15.0 cm version in 10 of 10. A miss costs about 420 points (400 cm bottle penalty plus no bonus).
Separate investigation observations:
Lateral error at bottle: Up to ±2.0 cm in 20 runs.
Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
Pusher B: 11.0 cm opening with V entry: 20 of 20 captures.

A. The Bottle Distance is set to 400.0 cm

B. A -20 bonus is still earned

C. The run is restarted

D. Nothing, if it crosses the line

Answer: _____

4. (2 points) Mounting the encoder on the motor shaft instead of the wheel axle with a 5:1 reduction:

A. Gives five times as many counts per wheel turn

B. Gives one-fifth as many counts

C. Has no effect

D. Prevents reversing

Answer: _____

5. (4 points) Why is a centimeter of vehicle error more costly than a centimeter of bottle error?

Answer: _____
_____
_____

6. (4 points) Why are large capacitors banned even though they store electrical energy?

Answer: _____
_____
_____

7. (4 points) Explain how overshoot can cost points.

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

Answer: _____
_____
_____

8. (4 points) Explain the tradeoff in choosing how far past the line to push the bottle.

Answer: _____
_____
_____

9. (4 points) Name two ways a run becomes a Failed Run.

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

Answer: _____
_____
_____

10. (4 points) Why should teams test with dented or off-weight bottles?

Unit: The bottle pusher: catching, pushing, and the narrow-opening bonus.
Worked-case data: Compare pushers with openings of 9.0 cm and 15.0 cm. The 9.0 cm version catches the bottle in 8 of 10 tests; the 15.0 cm version in 10 of 10. A miss costs about 420 points (400 cm bottle penalty plus no bonus).
Separate investigation observations:
Lateral error at bottle: Up to ±2.0 cm in 20 runs.
Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
Pusher B: 11.0 cm opening with V entry: 20 of 20 captures.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A 6.5 cm wheel drives a 100-count-per-revolution encoder through a 3:1 motor-to-wheel gear ratio, with the encoder on the motor shaft. How many counts for 8.60 m?

Answer: _____
_____
_____

12. (2 points) Which run scored lower (better)?

Supplied synthetic observations:
Pusher: Widest opening 20.0 cm.
Run 1: MP 6.0 cm off, bottle 5.0 cm past the line, time error 3.0 s.
Run 2: MP 3.0 cm off, bottle 8.0 cm short of the line, time error 0.5 s.

A. Run 1

B. Run 2

C. They tie

Answer: _____

13. (2 points) What is the key test for whether a component breaks the energy-storage rule?

Supplied synthetic observations:
Parts list: 8 x AA alkaline, Arduino, encoder, 12 V boost converter.
Inspection question: Does the boost converter store energy to exceed battery power?
Team explanation: It only steps up voltage continuously from the batteries; no large storage.

A. Whether it is electronic

B. Whether it changes voltage

C. Whether it lets the vehicle exceed the instantaneous energy available from the batteries

Answer: _____

14. (2 points) Define “Open-loop control” in the context of Electric Vehicle.

Answer: _____
_____
_____

15. (2 points) Define “Bottle Line Distance” in the context of Electric Vehicle.

Answer: _____
_____
_____

16. (2 points) Define “Event time” in the context of Electric Vehicle.

Answer: _____
_____
_____

17. (2 points) Define “Funnel (V) entry” in the context of Electric Vehicle.

Answer: _____
_____
_____

18. (2 points) Define “Gear ratio” in the context of Electric Vehicle.

Answer: _____
_____
_____